Wildfire3D: 3-D Wildfire Scene Reconstruction Using Multiview Aerial Imagery

Maoqin Jia, Xue Rui, Jun Zhang, and Weiguo Song

***Abstract*—Global climate change has increased the frequency of wildfires, making rapid and accurate situational awareness critical for effective firefighting. Although UAV-based imaging enables 2-D fire detection, localization, and tracking, the lack of reliable 3-D perception fundamentally limits firefighting precision. Recent advances in real-time 3-D reconstruction, such as 3-D Gaussian splatting (3DGS), offer new opportunities for wildfire monitoring; however, their effectiveness depends on reliable structure-from-motion (SfM) initialization, which can be challenging in wildfire scenes. Wildfire scenes pose severe challenges to SfM, including repetitive vegetation textures, dynamic flame-induced illumination variations, and frequent occlusions, which jointly cause ambiguous feature correspondences, missing keypoints, and unstable camera pose estimation, ultimately degrading reconstruction quality. In this work, we identify unstable SfM initialization as a key bottleneck that fundamentally limits 3-D reconstruction in wildfire environments. To address this issue, we propose Wildfire3D, a wildfire-aware 3-D reconstruction framework whose core contribution is a tailored sparse reconstruction strategy that enables robust SfM initialization under severe wildfire-specific visual interference. The proposed framework explicitly incorporates viewpoint-aware image pair selection, complementary feature matching, and spatial consistency-guided optimization to mitigate wildfire-induced SfM failure modes, thereby stabilizing camera pose estimation and reliably supporting downstream 3DGS reconstruction. In addition, we introduce Wildfire3Data, the first open-access multiview dataset specifically designed for wildfire 3-D reconstruction, providing a benchmark for evaluating reconstruction robustness in complex fire scenarios. Extensive experiments on five challenging wildfire scenes demonstrate that Wildfire3D consistently produces stable and accurate 3-D reconstructions, outperforming state-of-the-art (SOTA) wild scene reconstruction methods by stabilizing SfM initialization and achieving a 7.3% improvement in PSNR, a 5.5% gain in SSIM, and a 42.8% reduction in LPIPS.**

***Index Terms*—3-D Gaussian splatting (3DGS), 3-D reconstruction, fire monitoring, structure-from-motion (SfM), wildfire.**

I. Introduction

In RECENT years, global climate warming has triggered a period of frequent wildfires [1]. These fires often break out in remote forest regions with poor transportation access, leading to severe delays in firefighting response. Studies indicate that rapid intervention within the first hour of ignition can reduce the final burned area by more than around 80% [2], underscoring the urgent need for early and accurate fire environment awareness. Uncrewed aerial vehicles (UAVs), thanks to their mobility and safety, have emerged as a vital tool for wildfire monitoring. Existing research has demonstrated UAV-based 2-D image analysis tasks such as fire detection [3], fire localization [4], [5], and fire-front tracking [6] by leveraging visible or thermal infrared sensors combined with deep learning techniques. However, a significant gap remains between wildfire perception in 2-D images and in 3-D space. Compared with 2-D image analysis, 3-D reconstruction provides explicit spatial and geometric information that is essential for understanding wildfire behavior. Key wildfire characteristics, such as fire spread direction, flame height, and the spatial interaction between fire, vegetation, and terrain, are inherently 3-D and difficult to capture with purely 2-D representations, especially under complex conditions involving dense smoke, dynamic flames, and large viewpoint variations [7]. Reliable 3-D reconstruction enables a more comprehensive representation of wildfire scenes and supports downstream applications, including fireline localization, situational assessment, and UAV-assisted monitoring and decision support [8]. Therefore, robust 3-D scene reconstruction is not only a fundamental problem in computer vision and remote sensing but also a critical component for practical wildfire monitoring and firefighting operations.

The 3-D reconstruction methods play a critical role in firefighting operations by providing intuitive spatial information for fire monitoring and emergency response. Traditional 3-D reconstruction methods primarily relied on photogrammetry, including structure-from-motion (SfM) for sparse point cloud generation and multiview stereo (MVS) for densification [9], [10]. Later, deep learning approaches such as neural radiance fields (NeRFs) [11] introduced continuous volumetric representations for novel view synthesis. Recently, 3-D Gaussian splatting (3DGS) [12] emerged as a real-time alternative to NeRF, offering faster convergence and high rendering efficiency with comparable visual quality. It should be emphasized that the quality of 3DGS reconstruction critically depends on SfM-based initialization, especially the input camera poses and sparse geometry, which are typically derived from SfM pipelines.

However, wildfire environments present severe challenges for SfM-based initialization of 3DGS: high visual similarity in vegetation, heavy occlusions, and drastic lighting variations

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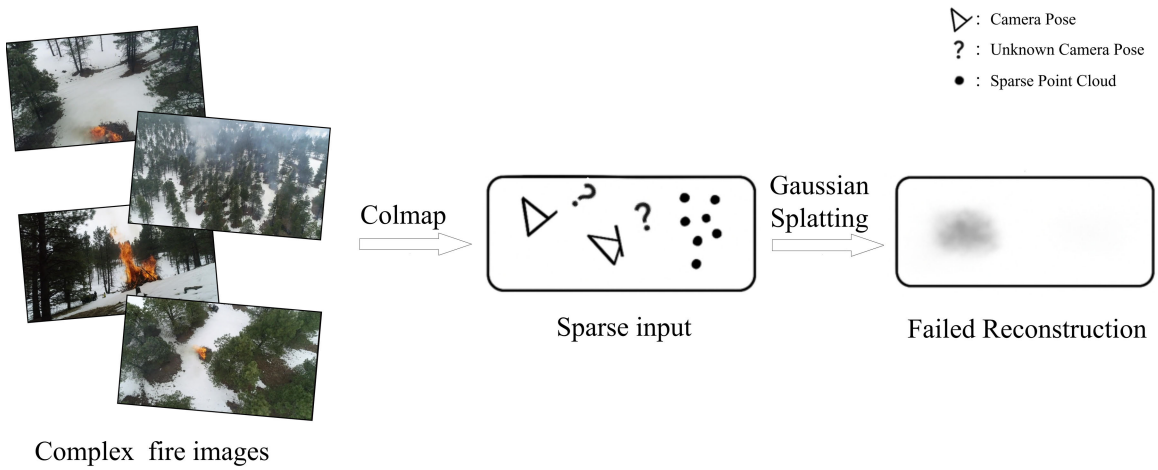


Fig. 1. Challenges of 3-D reconstruction in wildfire scenarios. Repetitive vegetation textures, dynamic flame illumination, severe occlusions, and large UAV viewpoint differences lead to unstable sparse inputs, ultimately causing failed wildfire scene reconstruction.

significantly weaken feature matching stability, leading to unreliable camera pose estimation and sparse reconstructions. Dynamic flames and limited UAV observation angles further exacerbate data sparsity and coverage limitations. Moreover, traditional SfM pipelines exhibit low automation and insufficient real-time performance, making them unsuitable for rapidly evolving fire conditions [13]. As shown in Fig. 1, conventional SfM frameworks (e.g., COLMAP [10]) fail in feature matching under dynamic disturbances such as smoke and flames. Combined with sparse inputs, these issues ultimately lead to poor 3-D reconstruction quality.

To address these challenges, we propose Wildfire3D, a 3-D reconstruction method for wildfire scenes. Specifically, we introduce a wildfire-specific SfM optimization pipeline designed to provide robust initialization for 3DGS. Unlike conventional SfM frameworks that fail under severe occlusion, repetitive vegetation textures, and dynamic illumination variations, Wildfire3D introduces viewpoint-aware image pair retrieval, dual-path feature extraction and matching, and spatial consistency fusion to enhance matching reliability and geometric accuracy. By systematically improving camera pose estimation and sparse point cloud quality, our pipeline ensures stable and accurate 3-D reconstruction in wildfire environments, which is essential for time-critical fire monitoring and emergency response.

The main contributions of this work are as follows.

1) We build the first **Wildfire 3D** reconstruction d**ata**set, (**Wildfire3Data**), encompassing diverse and complex wildfire scenarios, which can serve as a benchmark for this task.
2) We present the **Wildfire** scene **3D** reconstruction method (**Wildfire3D**) using multiview aerial images, which integrates viewpoint-aware image retrieval, dual-path feature matching, and spatial consistency fusion, effectively improving the robustness and accuracy of 3DGS reconstruction under complex wildfire scene.
3) Our method outperforms state-of-the-art (SOTA) outdoor scene reconstruction techniques across five representative wildfire scenarios, achieving mean improvements of 7.3% in PSNR, 5.5% in SSIM, and a reduction of 42.8% in LPIPS.

II. Related Work

A. UAV-Based Wildfire Monitoring and Its Limitations

UAVs have become an essential component of modern wildfire management due to their high mobility, wide operational coverage, cost-effectiveness, and ability to operate safely in hazardous environments. They are extensively employed for fire detection, confirmation, and continuous monitoring tasks [4], [5]. However, most existing methods primarily rely on 2-D image, which is insufficient for capturing the 3-D spatial characteristics of fire scenes, thus limiting comprehensive fire environment perception. To address the need for 3-D modeling, early work on UAV-based 3-D reconstruction for wildfires includes acquiring high-resolution fire imagery using UAV-mounted multispectral cameras and performing 3-D modeling with GIS techniques. For instance, Sargento et al. [14] proposed a wildfire georeferencing method based on SfM and iterative closest point (ICP) algorithms for mid-altitude wildfire monitoring, demonstrating the ability to reconstruct surrounding environments. Similarly, Ciullo et al. [15] introduced a multimodal stereo vision framework for UAVs, leveraging collaborative observation from visible and infrared cameras to estimate geometric properties of distant fire scenes, improving both accuracy and adaptability. Moreover, Sherstjuk et al. [16] addressed uncertainty in monitoring data by proposing a 3-D reconstruction method for fire fronts using UAV remote sensing data, capturing scene geometry through synchronized multicamera setups to enhance reconstruction reliability. Despite these advances, UAV-based wildfire 3-D reconstruction remains in its infancy and faces major challenges such as unreliable feature matching under repetitive vegetation textures, severe occlusions from smoke and flames, dynamic illumination variations, and large viewpoint differences. These limitations significantly undermine the accuracy and robustness of existing SfM pipelines in real wildfire conditions, motivating the need for specialized solutions.

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B. 3-D Reconstruction Methods and Challenges in Wildfire Scenarios

Current research trends indicate a transition from traditional geometric-based methods to deep learning-driven approaches, with growing emphasis on neural representations to enhance reconstruction accuracy and efficiency [17], [18], [19], [20]. Early methods relied primarily on multiview geometry techniques [9], later evolving into NeRF-based models [11] that deliver improved novel view synthesis. However, NeRF-based approaches remain computationally expensive and struggle to meet real-time application requirements. To address these limitations, Kerbl et al. [12] introduced 3DGS in 2023, an explicit scene representation method that models 3-D points as Gaussian functions and performs differentiable rendering without requiring neural network training. This approach offers efficient multiview synthesis and real-time 3-D rendering, making it highly suitable for time-critical applications. Subsequent works have extended 3DGS to dynamic scenes and large-scale environments, such as dynamic Gaussian graph modeling [21], few-shot reconstruction [22], and partition-based parallel optimization for large scenes [23]. Moreover, sensor fusion approaches, including LiDAR–inertial–visual (LIV) integration [24], and stereo-based pipelines such as Dust3R [25], have been proposed to enhance robustness in complex environments.

Despite these advances, most existing research focuses on downstream improvements in rendering quality or large-scale optimization, while the upstream stage—camera pose estimation and sparse reconstruction—remains a critical bottleneck. Since 3DGS performance strongly depends on the accuracy of SfM initialization, unreliable feature matching and pose estimation under wildfire-specific conditions (occlusion, repetitive textures, and dynamic lighting) severely degrade reconstruction quality. Although recent feature matching techniques, such as AdaLAM [26] and Superglue [27], improve correspondence accuracy by leveraging geometric consistency and graph neural networks, they have not been systematically adapted to the challenging wildfire imagery domain. This motivates the development of wildfire-specific robust SfM pipelines to provide stable initialization for 3DGS and similar real-time reconstruction frameworks.

III. Method

Accurate 3-D reconstruction in wildfire scenarios is particularly challenging due to severe occlusions, repetitive vegetation textures, and drastic illumination variations caused by flames and smoke. These factors significantly degrade feature detection and matching reliability, leading to unstable camera pose estimation and frequent failure in 3DGS initialization. To address these challenges, our method introduces a wildfire-specific sparse reconstruction optimization pipeline that introduces targeted improvements at multiple stages of the SfM process. Unlike conventional SfM pipelines, our method incorporates two core innovations: 1) viewpoint-aware image retrieval to reduce large-baseline mismatches, and dual-path feature extraction and matching tailored for heterogeneous wildfire conditions and 2) spatial consistency fusion to ensure geometric reliability. The primary objective is to overcome wildfire-specific visual challenges such as severe occlusion, repetitive vegetation textures, and drastic illumination variations through more robust camera pose estimation, which in turn enables reliable 3DGS initialization and accurate reconstruction in highly dynamic fire environments.

As illustrated in Fig. 2, the proposed method comprises five core modules: image retrieval, dual-path feature extraction, dual-path feature matching, matching integration, and spatial position estimation. The design and motivation of each component are detailed as follows.

A. Wildfire3Data

Currently, there is no open-access dataset specifically designed for 3-D reconstruction in wildfire scenarios. To address this gap, we constructed a dataset tailored for wildfire reconstruction tasks by integrating the publicly available FLAME dataset [3] with wildfire scene data experimentally collected, following methodologies adopted in [28], and [29]. Hereafter, we refer to the constructed dataset as **Wildfire3Data**, which is specifically designed for wildfire-oriented 3-D reconstruction tasks. The FLAME dataset includes professionally curated wildfire imagery, comprising videos of single-fire and multifire scenes, with specifications summarized in Table I. During preprocessing, looping videos were segmented into frames using FFmpeg, extracting keyframes at fixed 0.2-s intervals to minimize motion blur. After screening, 44 high-quality samples were retained for single-fire scenes and 48 for multifire scenes. To enhance data diversity, we conducted additional UAV experiments to collect multiview imagery from three representative wildfire environments (nighttime, withered trees, and dense foliage), with details provided in Table I. These video sequences underwent the same extraction and filtering pipeline to ensure high-quality frame selection. For dataset partitioning, we selected every first frame for evaluation from every eight-frame sequence and the remaining frames for training. This strategy ensures both diversity and representativeness, providing a robust benchmark for evaluating wildfire-specific 3-D reconstruction methods.

Wildfire3Data is intended to serve as a reusable benchmark for wildfire-oriented 3-D reconstruction research. The FLAME subset follows its original open-access license, while the experimentally collected wildfire data will be released for academic research upon acceptance or made available to the research community upon reasonable request.

B. Image Retrieval

Aerial images captured for wildfire monitoring often exhibit highly variable viewpoints, where large-baseline differences can cause feature matching failures. To mitigate this, we adopt a convolutional neural network EfficientNet-B7 [30] to extract global image descriptors and compute image similarities. Considering practical constraints in wildfire applications, we define a parameter N: when the total number of images is less than N, an exhaustive search strategy is applied to generate all possible image pairs. Otherwise, EfficientNet-B7 is used to compute Euclidean distances between global descriptors,

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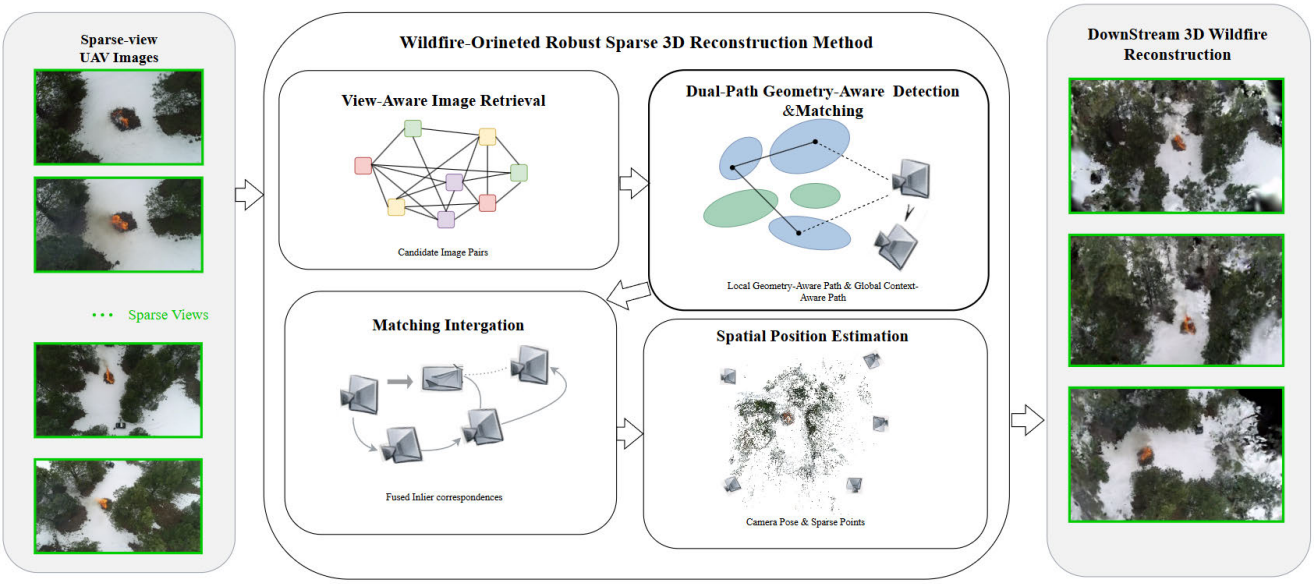


Fig. 2. Overview of the proposed Wildfire3D. The framework stabilizes structure-from-motion in wildfire scenes with occlusions and appearance changes through four main modules: **view-aware image retrieval**, **dual-path geometry-aware detection and matching**, **matching integration**, and **spatial position estimation**. Together, these modules produce geometrically reliable correspondences, enabling robust camera pose estimation and sparse structure recovery from limited-view UAV imagery. The recovered poses and sparse structure further support downstream 3DGS for wildfire scene reconstruction.

TABLE I

DATASET DETAILS FOR 3-D RECONSTRUCTION IN WILDFIRE SCENARIOS

Data Source	Data Type	Quantity	Resolution
FLAME [3]	Single-fire scene panoramic images	44	1280×720
	Multi-fire scene panoramic images	48	1280×720
Experimental Data	Nighttime scene panoramic images	32	960×540
	Withered tree scene multi-view images	57	1000×750
	Dense foliage scene panoramic images	33	1920×1080

and candidate image pairs are selected based on a similarity threshold derived from the distance matrix. Unless otherwise specified, N is set to 30 in all experiments.

C. Feature Extraction

Wildfire imagery often exhibits occlusions, repetitive textures, and substantial illumination variations, resulting in pronounced appearance differences across image regions. These complexities hinder the ability of conventional single-path keypoint detection methods to simultaneously ensure robustness and spatial coverage. Therefore, we propose a dual-path feature extraction and matching strategy that integrates two complementary methods to improve detection stability under wildfire conditions. The first path employs FireKeyNet for keypoint detection and AdaLAM for matching. FireKeyNet demonstrates strong geometric invariance and highly discriminative descriptors, making it suitable for structurally distinct regions with significant viewpoint changes. The second path leverages the lightweight ALIKED detector coupled with Superglue for end-to-end matching, providing superior performance in blurred or low-texture regions while producing denser and more uniformly distributed keypoints. These two detectors exhibit complementary strengths: FireKeyNet excels in geometrically prominent local regions, whereas ALIKED effectively handles weak-texture or degraded areas. Their synergy significantly improves overall spatial coverage and robustness, forming a solid foundation for subsequent feature matching and pose estimation.

1) FireKeyNet: Inspired by recent advances in keypoint detection [31], we design FireKeyNet, a feature detection network tailored to local regions with distinctive geometric structures such as flame and smoke boundaries. FireKeyNet consists of three key components. First, an orientation estimation module employs the Orinet algorithm [32] to estimate keypoint angles. Second, the keypoint detection module integrates KeyNet [33], SIFT [34], and the DoG detector [35] for multiscale localization, merging detected keypoints by retaining one representative point in spatially proximate positions and all nonoverlapping points. After merging, AffNet [36] is applied for affine shape estimation. Finally, the descriptor generation module employs HardNet [37] to produce highly discriminative descriptors.

AffNet [36] is a convolutional neural network designed to estimate local affine shapes, aiming to enhance the reliability and accuracy of image feature matching. Traditional feature detectors, such as SIFT, perform poorly under changes in illumination and viewpoint, failing to ensure reliable feature correspondences. AffNet learns affine shapes and orientations and introduces a novel hard negative-constant loss function, which combines the advantages of triplet loss and contrastive

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loss to optimize affine region estimation. The specific formulation is as follows:

$$L_{\text{HardNeg}} = \sum_{i=1}^{n} \max\left(0, 1 + d\left(s_i, \hat{s}_i\right) - d\left(s_i, N_i\right)\right). \quad (1)$$

Here, $d(s_i, \hat{s}_i)$ denotes the distance between the ith sample s_i and its corresponding positive sample $\hat{s}_i$, where smaller values indicate higher similarity between positive pairs; $d(s_i, N_i)$ denotes the distance between s_i and its hardest negative sample N_i, where larger values indicate greater dissimilarity between negative pairs. The core idea of this loss function is to encourage the model to learn more discriminative features by comparing the positive pair distance $d(s_i, \hat{s}_i)$ with the negative pair distance $d(s_i, N_i)$. A term contributes to the overall loss only when $1 + d(s_i, \hat{s}_i) - d(s_i, N_i) > 0$; otherwise, the contribution is zero.

KeyNet [33] combines handcrafted filters with learnable CNN filters, enabling efficient and stable keypoint detection within a shallow multiscale architecture. The handcrafted filters provide anchor structures, while the learnable CNN filters are responsible for localization, scoring, and ranking of keypoints. The network extracts image keypoints at different levels through a scale-space representation.

In this net, HardNet provides compact and discriminative 128-D descriptors, while AffNet improves correspondence robustness by normalizing local affine deformations. By focusing on geometrically distinctive structures such as flame boundaries, smoke edges, and structured vegetation, FireKeyNet yields stable and reliable keypoints under significant viewpoint and illumination changes commonly observed in wildfire imagery.

2) ALIKED: To complement the geometrically stable but relatively sparse keypoints extracted by FireKeyNet, we adopt ALIKED [38] as a lightweight detector–descriptor for improving keypoint coverage in degraded wildfire imagery. Wildfire scenes frequently contain blurred regions, low-texture backgrounds, and reduced contrast caused by smoke and illumination fluctuations, where conventional detectors tend to fail. ALIKED is specifically designed to handle such challenging conditions by producing dense and uniformly distributed keypoints with reliable descriptors, even in weakly textured or motion-blurred regions. In our dual-path setting, ALIKED serves as a complementary branch that enhances the spatial completeness of detected features, ensuring the presence of sufficient correspondences in regions where geometry-driven detectors are less effective. In this work, we directly employ the pretrained ALIKED model [38] without additional fine-tuning. The resulting keypoints and descriptors provide a robust basis for subsequent feature matching and cross-path integration under complex wildfire-induced appearance variations.

D. Feature Matching

In complex natural scenarios such as wildfires, images often suffer from illumination changes, occlusions, and motion blur, which prevent traditional nearest neighbor matching strategies based on feature descriptors from producing stable and accurate correspondences. Therefore, this study introduces two complementary feature matching mechanisms: the AdaLAM algorithm, which enforces geometric consistency, and the Superglue method, which uses deep learning for matching.

1) AdaLAM: Initial feature correspondences obtained from descriptor similarity often contain a large number of outliers in wildfire scenes due to severe occlusions, repetitive textures, and significant viewpoint variations. Such conditions make global geometric model fitting unreliable at early matching stages. To address this issue, we adopt adaptive locally affine matching (AdaLAM) [26] as a local geometric verification mechanism. AdaLAM verifies feature correspondences based on local affine consistency without assuming a single global geometric model, which is particularly suitable for highly cluttered and partially occluded wildfire environments. Specifically, for a seed correspondence (p_i, p_i'), AdaLAM fits a local affine transformation within its spatial neighborhood $\mathcal{N}(p_i)$

$$p_j' \approx A p_j + t \quad (2)$$

where $A \in \mathbb{R}^{2\times 2}$ and $t \in \mathbb{R}^2$ denote the local linear transformation and translation, respectively. Correspondences that are inconsistent with the local affine model are treated as outliers. By performing local geometric verification in parallel across multiple image regions, AdaLAM effectively filters mismatches caused by occlusion and large-baseline variations, yielding a set of geometrically consistent correspondences without relying on global model estimation. This property makes AdaLAM well suited for wildfire scenarios, where only locally reliable structures may be observable.

2) Superglue: In the traditional feature correspondence pipeline, matching between descriptors typically relies on Euclidean distance and nearest neighbor search strategies. However, during image acquisition in wildfire scenarios, complex environmental conditions such as uneven illumination and motion blur often occur, causing static local feature-based matching methods to produce numerous mismatches. In particular, without modeling contextual structural information, it is difficult to ensure the global consistency and robustness of correspondences.

To address this, this study introduces the Superglue algorithm [27], an end-to-end learnable feature matching method based on graph neural networks. Unlike traditional methods that rely solely on descriptor similarity, Superglue explicitly constructs a graph structure among keypoints, integrating spatial positions, descriptor features, and interpoint relationships, and uses a Transformer module to perform global context modeling, thereby achieving more reliable estimation of feature correspondences.

Its network architecture consists of two main stages. First, the feature sets of the two images are processed through multiple layers of self-attention and cross-attention mechanisms to enhance structural relationships among local features. Subsequently, a Sinkhorn layer [39] is applied to all possible point pairs for bi-stochastic normalization, producing a soft matching probability matrix

$$M = \text{Sinkhorn}(S), \quad S_{ij} = \left\langle \hat{d}_i, \hat{d}_j' \right\rangle \quad (3)$$

where S denotes the enhanced descriptor similarity matrix and M represents the final soft matching probability matrix. Based

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on this, the point pairs with the highest probability are selected as the final matches, effectively eliminating low-confidence mismatches.

E. Matching Integration

While the two matching paths above handle complementary failure modes in wildfire imagery, relying on either local or global matching alone is still insufficient for robust correspondences under severe occlusion and appearance degradation. We therefore propose a dual-path integration strategy that merges locally verified and globally consistent matches into a single reliable set. Specifically, our framework combines FireKeyNet with AdaLAM and ALIKED with Superglue, improving matching accuracy while remaining robust.

Specifically, the two paths provide complementary properties, with FireKeyNet-AdaLAM emphasizing geometric stability and ALIKED-Superglue enhancing spatial coverage and global consistency.

At the matching level, AdaLAM focuses on enforcing local geometric consistency, whereas Superglue emphasizes global structural coherence, resulting in two complementary correspondence sets with different reliability characteristics.

Given the two independently verified correspondence sets produced by the FireKeyNet-AdaLAM and ALIKED-Superglue paths, we perform correspondence-level fusion based on spatial consistency. Specifically, for matches originating from different paths, if their 2-D locations in both images lie within a predefined proximity threshold, they are treated as observations of the same underlying correspondence and merged into a single match. In this work, the proximity threshold is set to 0.3 normalized pixel distance. All nonoverlapping correspondences are retained to preserve diversity and spatial coverage. Through this fusion process, the integrated matching set combines the geometric reliability of locally verified matches with the spatial completeness and global coherence of context-aware matches, yielding a robust correspondence set for subsequent geometric verification and camera pose estimation.

F. Spatial Position Estimation

Given the fused 2-D–2-D correspondences obtained from the matching integration stage, this section estimates relative camera poses and reconstructs sparse 3-D geometry through robust geometric verification and incremental SfM.

To verify geometric consistency between image pairs, we employ the USAC_MAGSAC method [40], [41] to estimate the fundamental matrix $\mathbf{F}$ under severe outliers. The fundamental matrix describes the epipolar geometry between two images, and for true matching point pairs $(\mathbf{x}, \mathbf{x}')$, the following epipolar constraint should hold:

$$\mathbf{x}'^{\top}\mathbf{F}\mathbf{x} = 0 \tag{4}$$

where $\mathbf{x}$ and $\mathbf{x}'$ represent the homogeneous pixel coordinates of the same spatial point in the two images. This constraint reflects the geometric consistency that the projections of the same 3-D point must satisfy across two views.

USAC_MAGSAC combines the error evaluation mechanism of MAGSAC [41] with the USAC [40] framework, enabling more stable inlier identification and parameter estimation without relying on an initial inlier threshold. This is particularly suitable for complex conditions in fire scene images, such as severe occlusion and significant scale variations. After completing geometric verification of the matching point sets, we use the incremental SfM module provided by pyCOLMAP to perform relative camera pose estimation and sparse point cloud generation. Initially, verified image pairs are selected for initialization, and an initial 3-D structure is constructed using triangulation. Subsequently, the remaining images are incrementally incorporated into the reconstruction process, continually expanding the sparse model through incremental pose estimation and triangulation.

During this process, a global optimization strategy minimizing reprojection error, known as bundle adjustment (BA), is introduced. Its optimization objective is

$$\min_{\mathbf{R}_i,\mathbf{t}_i,\mathbf{X}_j} \sum_{i,j} \left\| \mathbf{x}_{ij} - \pi\left(\mathbf{R}_i\mathbf{X}_j + \mathbf{t}_i\right) \right\|^2 \tag{5}$$

where $\mathbf{x}_{ij}$ denotes the observed position of 3-D point $\mathbf{X}_j$ in the ith image; $\mathbf{R}_i$ and $\mathbf{t}_i$ are the camera rotation matrix and translation vector, respectively; and $\pi(\cdot)$ is the camera projection function. By minimizing the reprojection error across all observations, the system obtains higher-precision sparse 3-D structures and camera extrinsic estimates.

The spatial position estimation method employed in this section integrates a robust fundamental matrix estimation strategy with an industrial-grade SfM reconstruction framework, effectively recovering the spatial relationships among multiview images in complex fire scene environments. This process provides accurate camera poses and sparse point clouds for subsequent 3-D reconstruction.

G. 3-D Reconstruction

Based on the estimated camera poses and sparse point cloud obtained from the preceding pipeline, this section adopts 3DGS [12] as a downstream reconstruction module for explicit 3-D modeling of wildfire scenes. Specifically, anisotropic Gaussian ellipsoids are initialized from the SfM-derived camera poses and sparse geometry and projected onto the image plane via camera intrinsics to enable efficient differentiable rendering. Following the standard 3DGS formulation, the Gaussian parameters—including position, shape, color, and opacity—are optimized in an end-to-end manner using a multiview photometric consistency objective, jointly modeling the geometric structure and radiometric appearance of the scene

$$\min_{\Theta} \sum_{i} \mathcal{L}_{\text{photo}}\left(\hat{I}_i\left(\Theta\right), I_i\right) \tag{6}$$

where Θ denotes the set of all Gaussian parameters, $\hat{I}_i(\Theta)$ is the image rendered from the ith viewpoint, I_i is the corresponding ground-truth image, and $\mathcal{L}_{\text{photo}}$ represents the pixel-wise reconstruction error. By leveraging the explicit geometric initialization provided by the proposed SfM pipeline, 3DGS enables fast rendering and high-quality novel view synthesis

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in challenging wildfire environments, supporting accurate 3-D reconstruction and subsequent scene analysis.

IV. Results

A. Experimental Setup

Extensive experiments were conducted on various wildfire datasets, including the FLAME [3] dataset and our own collected dataset. Specifically, the FLAME dataset provides single-fire and multifire scene images, while our experimentally collected dataset includes nighttime, withered tree, and dense foliage scenes. We evaluated the quality of image reconstruction across these five representative wildfire scenes, covering multiple types of wildfires. All training and inference were performed and tested on a single RTX3090 GPU to ensure a fair comparison. In addition to comparing with existing SOTA wildfire image reconstruction methods, we also conducted ablation studies on different components to validate their effectiveness.

B. Parameter Settings

This section describes the parameter settings and implementation details used in all experiments. In particular, we explain the image pairing strategy and the rationale behind the parameter N. To balance computational efficiency and matching robustness, we introduce a parameter N to control the image pairing strategy. When the number of input images is smaller than N, exhaustive pairing is performed to generate all possible image pairs. Otherwise, global image descriptors extracted by EfficientNet-B7 are used to compute image similarities, and only candidate image pairs with high similarity are retained for subsequent feature matching. In this work, N is set to 30. When the number of images is below this threshold, the total number of image pairs remains limited and the computational overhead of exhaustive matching is relatively low. As the image number increases, the number of potential image pairs grows quadratically, while many pairs exhibit weak or no visual overlap, especially in wildfire scenarios affected by smoke, dynamic flames, and rapid viewpoint changes. Under such conditions, retrieval-based pairing effectively filters out irrelevant image pairs, significantly reducing unnecessary matching operations. Empirically, we observed that moderate variations of N around this value lead to similar reconstruction performance, indicating that the proposed method is not sensitive to the exact choice of N. Therefore, $N = 30$ is adopted as a balanced tradeoff between efficiency and robustness in all experiments.

C. Evaluation Metrics

For the task of image reconstruction in wildfire scenarios, we adopt three complementary evaluation metrics to quantify reconstruction quality from different perspectives: peak signal-to-noise ratio (PSNR), structural similarity index (SSIM), and learned perceptual image patch similarity (LPIPS). The evaluation procedure is conducted in three stages: alignment of ground-truth and reconstructed images, computation of per-pixel error maps, and analysis in the deep feature space.

As a traditional image quality metric, PSNR quantifies the pixel-level difference between the reconstructed image I_{render} and the ground truth I_{gt} using the mean squared error (mse) and is particularly sensitive to high dynamic range regions such as flames

$$\text{PSNR}\left(I_{\text{gt}}, I_{\text{render}}\right) = 10\log_{10}\left(\frac{L^2}{\text{mse}}\right) \tag{7}$$

$$\text{mse} = \frac{1}{HW}\sum_{i=1}^{H}\sum_{j=1}^{W}\left|I_{\text{gt}}(i,j) - I_{\text{render}}(i,j)\right|_2^2 \tag{8}$$

where $L = 255$ is the maximum pixel value for 8-bit images and $H \times W$ denotes the image resolution. This metric effectively reflects the accuracy of flame brightness reconstruction.

Considering geometric deformations in wildfire scenes, SSIM analyzes local similarity in brightness (μ), contrast (σ), and structure (σ_{xy}) using 11×11 pixel windows

$$\text{SSIM}(x,y) = \frac{\left(2\mu_x\mu_y + C_1\right)\left(2\sigma_{xy} + C_2\right)}{\left(\mu_x^2 + \mu_y^2 + C_1\right)\left(\sigma_x^2 + \sigma_y^2 + C_2\right)} \tag{9}$$

where $C_1 = (0.01L)^2$ and $C_2 = (0.03L)^2$ are stability constants. Global SSIM is obtained by averaging over all windows, and it is particularly effective at detecting texture distortions caused by smoke occlusion.

To capture dynamic flame artifacts that are difficult to represent with traditional metrics, we employ the LPIPS metric, which is more sensitive than PSNR and SSIM to temporal artifacts such as flame flickering and smoke diffusion. Distances are computed in the feature space of the pretrained VGG-16 network's conv3-3 layer

$$\text{LPIPS} = \sum_{l\in\mathcal{L}}\frac{1}{H_l W_l}\sum_{h=1}^{H_l}\sum_{w=1}^{W_l}\left|\phi^l\left(I_{\text{gt}}\right)_{h,w} - \phi^l\left(I_{\text{render}}\right)_{h,w}\right|_2^2 \tag{10}$$

where $\mathcal{L}$ denotes the set of selected feature layers and $H_l \times W_l$ is the spatial size of the lth layer's feature map.

D. Compared Methods

To comprehensively evaluate the performance of our method, we selected three existing SOTA wild image reconstruction methods for comparison.

1) *Colmap-3DGS (3DGS)*: A traditional two-stage pipeline. First, COLMAP is used to compute accurate camera poses and sparse point clouds, which are then used to initialize and train the 3DGS [12] model. This pipeline relies on high-quality preprocessing and is often considered the performance upper bound for 3DGS.
2) *Gaussian-Wild (GS-Wild)*: An end-to-end framework that jointly optimizes camera poses and Gaussian scene representation for unconstrained, multisource, and sparsely viewed outdoor images [42], eliminating the dependency on high-quality SfM preprocessing. This approach significantly improves reconstruction stability and generalization in complex outdoor conditions with variable data quality and sparse viewpoints, making it an SOTA solution for challenging field scenes.
3) *ColmapFree-3DGS (CF-3DGS)*: An end-to-end, preprocessing-free approach [43]. It jointly optimizes

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TABLE II

QUANTITATIVE COMPARISON OF WILDFIRE IMAGE RECONSTRUCTION QUALITY USING PSNR (↑), SSIM (↑), AND LPIPS (↓). THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Scene	Colmap-3DGS			GS-Wild			ColmapFree-3DGS			FSGS			Ours		
	PSNR↑	SSIM↑	LPIPS↓	PSNR↑	SSIM↑	LPIPS↓	PSNR↑	SSIM↑	LPIPS↓	PSNR↑	SSIM↑	LPIPS↓	PSNR↑	SSIM↑	LPIPS↓
Single fire	26.034	0.823	0.274	25.833	0.842	0.368	12.055	0.455	0.584	26.086	0.825	0.317	**27.244**	**0.851**	**0.249**
Multi fire	18.024	0.662	0.354	**18.939**	**0.711**	0.458	9.761	0.298	0.631	17.020	0.623	0.451	18.909	0.676	**0.344**
Nighttime	23.862	0.852	0.340	**27.700**	**0.896**	0.409	20.395	0.756	0.420	26.334	0.875	0.334	26.313	0.884	**0.274**
Withered tree	19.920	0.444	0.537	19.901	0.468	0.752	16.464	0.241	0.596	19.357	0.447	0.506	**22.107**	**0.687**	**0.323**
Dense foliage	21.532	0.720	0.465	16.314	0.724	0.842	15.738	0.457	0.630	22.027	0.739	0.508	**22.047**	**0.740**	**0.431**
Mean	21.874	0.700	0.394	21.734	0.728	0.566	14.883	0.441	0.572	22.165	0.702	0.423	**23.324**	**0.768**	**0.324**

camera poses while training the 3DGS model, eliminating the need for SfM preprocessing and providing potential for real-time reconstruction. Its pose accuracy is constrained by online estimation, serving as a key baseline for evaluating real-time performance scenarios.

4) *FSGS*: A real-time Gaussian splatting method designed for few-shot view synthesis [22]. By simplifying Gaussian representations and accelerating optimization, FSGS enables fast reconstruction from a limited number of input views, making it a representative solutiuon for sparse-view 3-D Gaussian reconstruction.

E. Quantitative Results

Table II presents a quantitative comparison of reconstruction quality between the proposed method and recent 3-D reconstruction approaches using PSNR, SSIM, and LPIPS. As shown, Wildfire3D consistently enhances the capability and robustness of 3DGS [12] in wildfire scenarios and achieves superior reconstruction quality compared to existing methods, with stable performance across diverse fire scenes. Specifically, compared to the SOTA GS-Wild model, Wildfire3D achieves an average PSNR improvement of 7.3%, an average SSIM improvement of 5.5%, and an average LPIPS reduction of 42.8%.

In single-fire scenes, Wildfire3D achieves the best performance across all three evaluation metrics, demonstrating a stronger modeling capability for high-frequency flame details. In multifire scenes, although GS-Wild slightly outperforms Wildfire3D in terms of PSNR and SSIM, Wildfire3D achieves a substantial reduction in LPIPS (over 40%), indicating that its reconstructions are perceptually more natural.

Regarding environmental robustness, in low-light conditions such as nighttime scenes, Wildfire3D maintains high SSIM while significantly improving LPIPS, demonstrating that its light field modeling can effectively adapt to complex illumination and avoid overblurring or noise artifacts commonly observed in traditional methods under dark conditions. For complex background scenarios involving interactions between flames and withered trees, Wildfire3D exhibits notable improvements in both PSNR and SSIM, with SSIM increasing by more than 50%, indicating a stronger ability to model dynamic occlusions and complex geometric structures, thereby enabling more accurate reconstruction of flame–background interactions.

TABLE III

QUANTITATIVE COMPARISON OF RUNTIME PERFORMANCE FOR WILDFIRE IMAGE RECONSTRUCTION USING TIME (↓). THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Scene	Colmap-3DGS	GS-Wild	ColmapFree-3DGS	FSGS	Ours
Single fire	**3min44s**	20min20s	40min6s	7min33s	5min38s
Multi fire	**4min21s**	22min53s	42min23s	6min13s	5min58s
Nighttime	**2min03s**	12min56s	26min43s	3min59s	3min24s
Withered tree	**10min38s**	27min56s	43min32s	11min52s	11min18s
Dense foliage	**4min48s**	24min12s	50min11s	6min20s	6min52s
Mean	**5min7s**	21min39s	40min35s	7min11s	6min38s

In addition to reconstruction quality, the runtime performance of different methods is reported separately in Table III. The results show that Wildfire3D achieves these quality improvements while maintaining competitive computational efficiency, supporting its practical applicability in real-world wildfire monitoring scenarios.

F. Reconstruction Result

We present visual comparisons across five representative scenarios: singlefire, multifire, nighttime, withered tree, and dense foliage. The results are shown in Fig. 3. From the visual results, our method preserves finer details of fire and the surrounding environment, maintains more coherent scene structures, and produces visually plausible reconstructions even under challenging conditions such as low light, dense vegetation, or occlusions in the withered tree scene. Compared with 3DGS, CF-3DGS, GS-Wild, and FSGS, our results exhibit sharper edges, clearer textures, and fewer artifacts.

G. Ablation Study

To validate the effectiveness of the modules proposed in this study, we designed a series of ablation experiments. The results are shown in Table IV. Using Colmap-3DGS as the baseline, we evaluated the performance of two individual detector combinations: ALIKED-Superglue and FireKeyNet-AdaLAM. Results indicate that using the FireKeyNet-AdaLAM combination improves PSNR and SSIM by 4.8% and 9.1%, respectively, and reduces LPIPS by 17.0%, significantly outperforming the ALIKED-Superglue combination, which only achieves improvements of 2.4%, 5.9%, and

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Fig. 3. Qualitative comparison on different scenes. Comparisons were conducted with 3DGS, CF-3DGS, GS-Wild, and FSGS. The red and yellow boxes indicate two representative regions of interest, and the corresponding zoomed-in views are shown below each method for all scenes, highlighting sharper details and improved visual fidelity achieved by our approach.

TABLE IV

QUANTITATIVE EVALUATION ON WILDFIRE SCENE RECONSTRUCTION USING PSNR (↑), SSIM (↑), AND LPIPS (↓). ALL METHODS USE 3DGS WITH DIFFERENT SFM INITIALIZATIONS: STANDARD COLMAP, FIREKEYNET, ALIKED, AND OUR FULL FIRE-ADAPTED PIPELINE (OURS). THE BEST RESULTS ARE IN BOLD

Scene	Colmap-3DGS			Ours (w/o ALIKED)			Ours(w/o FireKeyNet)			Ours		
	PSNR↑	SSIM↑	LPIPS↓	PSNR↑	SSIM↑	LPIPS↓	PSNR↑	SSIM↑	LPIPS↓	PSNR↑	SSIM↑	LPIPS↓
Single fire	26.034	0.823	0.274	26.764	0.844	0.259	24.218	0.795	0.333	**27.244**	**0.851**	**0.249**
Multi fire	18.024	0.662	0.354	18.623	0.670	**0.341**	18.300	0.655	0.345	**18.909**	**0.676**	0.344
Nighttime	23.862	0.852	0.340	25.235	0.880	0.277	25.738	0.864	0.328	**26.313**	**0.884**	**0.274**
Withered tree	19.920	0.444	0.537	22.053	0.686	**0.322**	21.995	0.660	0.343	**22.107**	**0.687**	0.323
Dense foliage	21.532	0.720	0.465	21.939	0.738	0.438	21.744	0.731	0.455	**22.047**	**0.740**	**0.431**
Mean	21.874	0.700	0.394	22.923	0.764	0.327	22.400	0.741	0.361	**23.324**	**0.768**	**0.324**

8.4%. This validates the effectiveness of our wildfire-specific feature detector design. More importantly, the complete Wildfire3D framework achieves the best performance across all test scenes, with PSNR, SSIM, and LPIPS improvements of 6.6%, 9.7%, and 17.8%, respectively. Notably, it achieves the best LPIPS in single-fire scenes, maintains high SSIM under low-light conditions while significantly reducing perceptual artifacts, and substantially improves flame-object boundary

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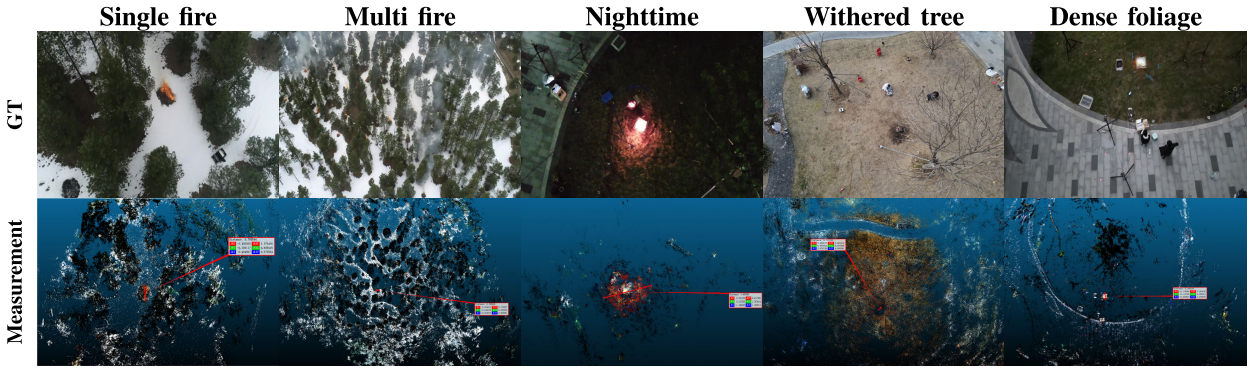


Fig. 4. Relative distance measurement within reconstructed wildfire point clouds across different scenarios. All images are rendered in identical display windows for fair visual comparison. The measurements are in COLMAP coordinates without absolute scale and can be converted to real-world units with geo-referenced camera poses.

TABLE V

QUANTITATIVE COMPARISON OF CAMERA POSE ESTIMATION ACROSS FIRE SCENE CATEGORIES USING MEAN REPROJECTION ERROR (↓). THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Method	Single fire	Multi fire	Nighttime	Withered tree	Dense foliage	Mean
Before optimization	0.427	**0.320**	0.430	0.306	0.658	0.428
After optimization	**0.398**	0.342	**0.405**	**0.272**	**0.647**	**0.413**

TABLE VI

EFFICIENCY ABLATION OF THE VIEWPOINT-AWARE IMAGE RETRIEVAL MODULE IN THE PROPOSED METHOD ACROSS DIFFERENT WILDFIRE SCENES (TIME ↓). THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Method	Single fire	Multi fire	Nighttime	Withered tree	Dense foliage	Mean
Time (w/o Retrieval)↓	5min51s	6min23s	3min26s	12min37s	7min43s	7min12s
Time (w/ Retrieval)↓	**5min38s**	**5min58s**	**3min24s**	**11min18s**	**6min52s**	**6min38s**

reconstruction in complex interaction scenes. Finally, as shown in Table V, the reprojection error is reduced after optimization across most scenarios, further demonstrating improved geometric accuracy and camera pose estimation. These results provide strong evidence for the necessity of the multidetector fusion strategy and support the superior performance of our method in 3-D reconstruction of wildfire scenarios.

In addition, we further analyze the impact of the image pairing strategy on reconstruction performance. Specifically, we compare exhaustive image pairing (w/o retrieval) with retrieval-based pairing using EfficientNet-B7 (w/retrieval) while keeping all other components of the pipeline unchanged. The corresponding results are also reported in Table VI. As shown in the table, introducing retrieval-based pairing substantially reduces the number of image pairs and matching cost, while maintaining comparable reconstruction quality across all wildfire scenes. This demonstrates that the proposed retrieval strategy effectively improves computational efficiency without degrading reconstruction accuracy.

H. Discussion on Evaluation Metrics for Wildfire Scenarios

While standard image-based metrics such as PSNR, SSIM, and LPIPS are widely adopted for evaluating novel view synthesis and 3-D reconstruction quality, we acknowledge that they are insufficient to fully capture application-oriented requirements in wildfire scenarios, such as flame volume, fire-front extent, or spread direction. Notably, the proposed method reconstructs explicit 3-D geometry in the form of sparse point clouds and 3-D Gaussian representations, which enables direct geometric inspection and spatial measurement. For example, as shown in Fig. 4, the distance between two points in the reconstructed point cloud can be measured, illustrating that fire-related spatial attributes (e.g., flame length and burned area extent) are, in principle, quantifiable.

However, in the current setting, the reconstructed geometry is defined in a COLMAP coordinate system without absolute scale, as the input imagery does not provide reliable GPS or metric camera pose information. We note that when accurate GPS or geo-referenced camera poses are available, the reconstructed 3-D scene can be aligned to a real-world coordinate system. In such cases, a conversion factor between COLMAP units and real-world units can be computed, enabling quantitative measurement of fire severity indicators such as flame length or burned area. We leave such application-level metric design and evaluation as an important direction for future work.

V. CONCLUSION

In this work, we introduced Wildfire3Data, the first open-access dataset dedicated to wildfire 3-D reconstruction and proposed a multistrategy sparse reconstruction pipeline tailored for complex fireground environments. By integrating image retrieval-based pair selection, dual-path feature extraction, complementary matching strategies, and spatial consistency fusion, our method significantly improves the robustness of camera pose estimation and sparse reconstruction under severe occlusions, repetitive textures, and illumination variations. Extensive experiments on multiple wildfire scenarios demonstrate that the proposed approach achieves consistent improvements over SOTA outdoor reconstruction methods in terms of PSNR, SSIM, and LPIPS, establishing a reliable solution for accurate 3DGS initialization in wildfire perception tasks.

Despite these advances, challenges remain in handling dynamic flames and rapidly moving smoke, where matching failures and error accumulation can still occur. Additionally, the current framework is limited by its reliance on sparse reconstruction and lack of semantic scene understanding. Future work will explore integrating semantic segmentation, thermal or multispectral imaging, and real-time reconstruction

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systems to further enhance structural awareness and stability in large-scale fire monitoring and emergency response.

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